SE375 SYSTEM PROGRAMMING SPRING 2025-2026
Laboratory Assignment

May 8, 2026

Objective

This lab introduces packet-level network analysis using Wireshark. Students will capture, filter, and interpret real network traffic. The lab is designed to span two sessions and includes both practical tasks and analytical questions.

Lab Tasks

Part 1: Packet Capture and Exploration

1. Open Wireshark and start capturing Ethernet traffic. There are various filters possible in Wireshark

2. Show the first N packets captured: As an example filter `frame.number <= 2` shows the first two.

3. By double-clicking on a frame, you can observe the content as ASCII.

4. By double-clicking on a frame, you can observe the packets also in HEX format.

5. After you have stopped capturing packets, you will be able to save the capture. Use 'File > Save As' for saving all capture data and 'File > Export Specified Packets' to save packets with filters applied.

6. You can later load a saved file to the Wireshark interface.

7. Find packets with a certain IP address using `ip.addr == x.x.x.x`. To match only the destination addresses, use `ip.dst == x.x.x.x` and to match the source addresses, `ip.src == x.x.x.x`.

8. Through 'View' > 'Time Display Format' you can designate the time format you want to see. The default is set to 'Seconds since Beginning of Capture'.

9. Find packets longer than N bytes. Example filter is `frame.len > 2048`

10. Similarly, you can find packets less than N bytes. `frame.len < 1024`

11. To find packets using only the specific protocols use protocol name in filter box. For example, for ARP simply type `arp` to filter box.

12. To examine all the packets going through a specific port use `tcp.port == X` for TCP ports and similarly `udp.port == X` for UDP.

13. An important thing is to be able to use multiple filters at the same time. You can use any Boolean logic to accomplish this with combinations of && and || and !. For example ip.dst == 192.168.0.15 && tcp.port == 22 filter finds packets that flow to IP address 192.168.0.15 through port number 22.

14. You can filter out packets you don't want. A filter is given as: !arp && !tcp What does this filter accomplish?

15. Now filter out all UDP packets going out from your IP address. You need to find your own IP address.

Questions

- How many packets were captured?
- Which protocol appears most frequently?
- What is the difference between ASCII and HEX views?
- Can you identify readable text in any packet?

Part 2: Filtering Techniques

1. Locate a TCP communication between two devices in your capture.
    - Apply the following filter: tcp
    - Find a packet that is part of a TCP connection.
    - Try to identify the connection establishment phase by locating packets with:
        - SYN
        - SYN-ACK
        - ACK

    Answer the following:

    - What are the source and destination IP addresses?
    - Which port numbers are used?
    - Can you observe the TCP handshake (SYN, SYN-ACK, ACK)?

2. Find a DNS request in your captured traffic.
    - Apply the following filter: udp.port == 53
    - Select one DNS packet.

    Answer the following:

    - What is the domain name being queried?
    - Which device sent the request?
    - Which server responded?

Submission

Submit your reports in DOCX or PDF format from the Blackboard system by using the template report document.